



**DURBAN UNIVERSITY OF TECHNOLOGY**  
**INYUVESI YASETHEKWINI YEZOBUCHWEPHESHE**

**FACULTY OF ACCOUNTING AND INFORMATICS**  
**DEPARTMENT OF INFORMATION TECHNOLOGY**

**2024**  
**YEAR END SUPPLEMENTARY EXAMINATION**

**INSTRUCTIONAL PROGRAMME :** ADVANCE DIPLOMA IN ICT

**INSTRUCTIONAL OFFERING :** SOFTWARE DEVELOPMENT AND MANAGEMENT

**PAPER NUMBER :** 1

**SUBJECT CODE/S :** SODM401/SFEN301

**DATE :** 2025/01/13

**DURATION :** 3 HOURS

**TIME :** 9:00 - 12:00

**TOTAL MARKS :** 100

**NUMBER OF PAGES :** 10 (Including cover page)

**EXAMINER/S :** Mr. N. Dlamini

**MODERATOR/S :** Mr. P. Motsilili

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**INSTRUCTIONS/REQUIREMENTS:-**

1. Ensure that your student number, name and surname appear on the answer sheet and MCQ card
2. Answer all questions in pen.
3. Use the mark allocation as a guide for the length of the answer
4. Section A: Circle only one answer on the MCQ card and use the answer booklet for Section B

***Do not turn the page until permission is given***

**SECTION A (40 Marks)**

**Answer all questions for this section by shading in your responses on the MCQ card provided.**

1. Which of the following statements is not true about form-based specifications?
  - a. Definition of the function or entity
  - b. Description of inputs and where they come from.
  - c. Information about the information needed for the computation and other entities used
  - d. Description of inputs and outputs
2. Requirements checking is a method that ensures the following except
  - a. Validity
  - b. Consistency
  - c. Predictability
  - d. Verifiability
3. Which validation technique involves developing tests for requirements to check testability?
  - a. Prototyping
  - b. Requirements reviews
  - c. Test-case generation
  - d. None of the above
4. Which of the following may not be the reason for changing requirements?
  - a. The business and technical environment of the system always changes after installation
  - b. The people who pay for a system and the users of that system are rarely the same people
  - c. The user community may have similar requirements and priorities that may be less conflicting or contradictory.
  - d. The final system requirements are inevitably a compromise between users
5. According to Mitch Kapo, a good software design must exhibit
  - a. Firmness
  - b. Commodity
  - c. Delight
  - d. All of the above
6. Which design model transforms analysis classes into design classes along with the data structures required to implement the software?
  - a. Component-level design
  - b. Interface design
  - c. Architectural design
  - d. Data/class design

13. \_\_\_\_\_ testing is a comprehensive and structured software testing process that evaluates a complete software application or system to ensure that it meets its intended requirements, functions correctly, and satisfies its specified goals
- Unit
  - System
  - Integrated
  - Performance
14. Biometrics or digital signatures, can be used as
- verification mechanisms
  - validation mechanism
  - testing mechanisms
  - All of the above
15. What is the purpose of a primary key in a relational database?
- To prevent data from being modified
  - To uniquely identify each record in a table
  - To establish a foreign key relationship
  - To define the data types of columns
16. Which software development methodology emphasizes continuous integration, automated testing, and short development cycles?
- Waterfall
  - Extreme Programming (XP)
  - Kanban
  - Scrum
17. \_\_\_\_\_ uses both validation and verification process.
- Testing
  - Inspection
  - A and B
  - None of the above.
18. Which of the following is associated with natural language when writing up requirements?
- Lack of clarity
  - Requirements confusion
  - Requirements amalgamation
  - All of the above

24. Which of the following assist in establishing a relationship between two tables in object-oriented development?

- a. Primary key
- b. Foreign key
- c. Inheritance
- d. Polymorphism

25. Which of the following is not a universal management activity?

- a. Project planning
- b. Process management
- c. Risk management
- d. People management

26. Which of the following is not an intrinsic reward in motivating people at workplaces?

- a. Money
- b. Recognition
- c. Achievement
- d. Responsibility

27. Mentcare system is a system designed for use in clinics attended by patients suffering from mental health problems and records details of their consultations and conditions. As patients arrive at the clinic a medical receptionist who manages the patients' database can either register a new patient or view patients' personal information. Nurses and doctors can view and edit patient records however only doctors can set up consultations and treat patients. Health care managers and doctors can generate a report from the system. However, the health care manager has an additional role to view patients' personal information and to export statistics of patients recorded in the Mentcare system. Identify the notation in which the scenario was written.

- a. Structured natural language
- b. Natural language
- c. Descriptions languages
- d. All of the above

28. \_\_\_\_\_ holds that modules should be designed in such a way that the algorithms and local data contained within them are inaccessible to other modules.

- a. Structural properties
- b. Extra-functional properties
- c. Information hiding
- d. Software refactoring

38. In object-oriented development polymorphism refers to a characteristic that greatly increases the effort required to extend the design.

- a. True                                      b. False

39. Inheritance looks at how responsibilities of a superclass are immediately inherited by all subclasses.

- a. True                                      b. False

40. Organizational requirements specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.

- a. True                                      b. False

## SECTION B (60 Marks)

### Question 1 (10 Marks)

- 1.1 Outline any 4 guidelines that may be used when documenting requirements. (4 marks)
- 1.2 Explain two considerations when choosing a design model? (2 marks)
- 1.3 Briefly explain the importance of conducting the following in project planning. (6 marks)
  - Project scheduling
  - Monitoring and reporting
  - Risk analysis

### Question 2 (10 Marks)

- 2.1 Water fall model is one of the popular software processes that are mostly implemented. What are the **phases** that are involved in this model and state the **Main** drawback of this model? (8 marks)
- 2.2 In Software Engineering, there are different types of Software testing that ensure software quality. List any two types of software Testing that you are familiar with. (2 marks)

**Question 4 (10 marks)**

4.1 List 6 reasons for information hiding. (6 marks)

4.2 Briefly explain the following concepts of designing for functional dependencies:(4 marks)

Coupling

Cohesion

**Question 5 (20 Marks)**

X-University is facing the critical issue on elections of choosing a leader who will represent the SRC and preserve its rights with the threat of a criminal element attempting to influence the outcome of the election. Third year students from DUT, IT department are approached to intervene. DUT students are meant to build an online voting system using blockchain. The following question are based on the design phase.

5.1 Identify potential risks associated with the transition to a new software process model and the implementation of system improvements. (6 marks)

5.2 List two potential non-functional requirements. (2 marks)

5.3 Identify any four Key role players on this application and draw a use case diagram of how these players interact with the application. (12 marks)